THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 8 — MATHS CHAPTER TEST SET A

Chapter 13: **Algebra Play** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Draw neat figures wherever required.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. Think of a number n. Add 5, multiply by 2, then subtract 10. The result is: 1

(a) $2n + 5$ **(b)** $5n$ **(c)** $2n$ **(d)** $n + 5$

2. Think of a number n. Add 6, multiply by 2, then subtract 12. The result is: 1

(a) $n + 6$ **(b)** $2n$ **(c)** $2n + 6$ **(d)** $6n$

3. The two-digit number with tens digit 8 and ones digit 9 can be written as: 1

(a) 8×9 **(b)** $10 \times 9 + 8$ **(c)** $8 + 9$ **(d)** $10 \times 8 + 9$

4. The sum of 29 and its reverse 92 is: 1

(a) 99 **(b)** 11 **(c)** 121 **(d)** 132

5. In a number pyramid, each block is the sum of the two below it. The bottom row is 6, 6, 3. The top block is: 1

(a) 15 **(b)** 23 **(c)** 27 **(d)** 21

6. The sum of three consecutive integers is always divisible by: 1

(a) 4 **(b)** 2 **(c)** 3 **(d)** 6

7. A number is divisible by 9 if and only if the sum of its digits is divisible by: 1

(a) 9 **(b)** 6 **(c)** 3 **(d)** 12

8. In a calendar, the dates in a 2×2 block starting at 9 are 9, 10, 16, 17. Their sum is: 1

(a) 54 **(b)** 52 **(c)** 44 **(d)** 50

9. In a calendar, the dates in a 2×2 block starting at 40 are 40, 41, 47, 48. Their sum is: 1

(a) 178 **(b)** 168 **(c)** 176 **(d)** 174

10. For any integer n, $(n + 2)^2 - (n - 2)^2$ equals: 1

(a) $4n$ **(b)** 8 **(c)** $8n$ **(d)** 4

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 10–15 words each.

11. Show that the trick 'Think of a number, add 7, double, subtract 14' always gives double the number. 2

12. Write a two-digit number $\overline{ab}$ in algebraic form and its reverse. 2

13. Show that the sum of a two-digit number and its reverse is divisible by 11. 2

14. The bottom row of a number pyramid is 3, 5, 4. Find the top block. 2

SECTION — C (Short Answer Type) 4 × 3 = 12

Answer in about 30–40 words each.

15. Use a number pyramid with bottom row 3, 5, 7: find all rows and the top, and write the top in terms of the bottom numbers. 3

16. Use a number pyramid with bottom row 2, 9, 4: find all rows and the top, and write the top in terms of the bottom numbers. 3

17. Use a number pyramid with bottom row 6, 1, 8: find all rows and the top, and write the top in terms of the bottom numbers. 3

OR

Use a number pyramid with bottom row 4, 4, 9: find all rows and the top, and write the top in terms of the bottom numbers. 3

18. Use a number pyramid with bottom row 5, 8, 3: find all rows and the top, and write the top in terms of the bottom numbers. 3

OR

Use a number pyramid with bottom row 7, 2, 6: find all rows and the top, and write the top in terms of the bottom numbers. 3

SECTION — D (Long Answer Type) 2 × 5 = 10

Answer in about 80–100 words each.

19. (i) Show the number trick: 'Think of n, add 5, multiply by 3, subtract 15, divide by 3'. 5
(ii) What does the trick give?
(iii) Create a new trick giving 10 always.

OR

Case study: A teacher makes a number pyramid with bottom row a, b, c. 5
(i) Write rows 2 and 3.
(ii) If the bottom is 5, 2, 7, find the top.
(iii) What should b be for the top to be 30 if $a = 6$ and $c = 8$?

20. (i) Prove that the sum of the four dates of a 2×2 block of a calendar is $4x + 16$. 5
(ii) Can the sum be 100? 130?

OR

(i) Prove that the product of two numbers with fixed sum is largest when they are equal. 5
(ii) Find the largest area of a rectangle with perimeter 36 m.

END OF QUESTION PAPER Total: 40 Marks

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